

1.5 Inverse of a Matrix

Definition: If A is a square matrix, and if a matrix B of the same size can be found such that

$$AB = BA = I \text{ (Identity Matrix)}$$

then A is said to be **invertible** (or **nonsingular**) and B is called an **inverse** of A . If no such matrix B can be found, then A is said to be **singular**.

The relationship $AB = BA = I$ is not changed by interchanging A and B , so if A is invertible and B is an inverse of A , then it is also true that B is invertible, and A is an inverse of B . Thus, when $AB = BA = I$, we say that A and B are inverses of one another.

Let

$$A = \begin{bmatrix} 2 & -5 \\ -1 & 3 \end{bmatrix} \text{ and } B = \begin{bmatrix} 3 & 5 \\ 1 & 2 \end{bmatrix}$$

Then

$$AB = \begin{bmatrix} 2 & -5 \\ -1 & 3 \end{bmatrix} \begin{bmatrix} 3 & 5 \\ 1 & 2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I$$

$$BA = \begin{bmatrix} 3 & 5 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} 2 & -5 \\ -1 & 3 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I$$

Thus, A and B are invertible and each is an inverse of the other.

Remarks:

1. In general, a square matrix with a row or column of zeros is singular.
2. An invertible matrix has exactly one inverse.

Formula for the Inverse of a matrix:

If A is an $n \times n$ matrix and $\det(A) \neq 0$, then

$$A^{-1} = \frac{1}{\det(A)} (\text{adj } A) = \begin{bmatrix} \frac{A_{11}}{\det(A)} & \frac{A_{21}}{\det(A)} & \cdots & \frac{A_{n1}}{\det(A)} \\ \frac{A_{12}}{\det(A)} & \frac{A_{22}}{\det(A)} & \cdots & \frac{A_{n2}}{\det(A)} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{A_{1n}}{\det(A)} & \frac{A_{2n}}{\det(A)} & \cdots & \frac{A_{nn}}{\det(A)} \end{bmatrix}$$

For a 2x2 matrix this formula is easy to use. For example

The matrix

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

is invertible if and only if $ad - bc \neq 0$, in which case the inverse is given by the formula

$$A^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

Example 1:

In each part, determine whether the matrix is invertible. If so, find its inverse.

(a) $A = \begin{bmatrix} 6 & 1 \\ 5 & 2 \end{bmatrix}$

(b) $A = \begin{bmatrix} -1 & 2 \\ 3 & -6 \end{bmatrix}$

Solution

(a) The determinant of A is $\det(A) = (6)(2) - (1)(5) = 7$, which is nonzero. Thus, A is invertible, and its inverse is

$$A^{-1} = \frac{1}{7} \begin{bmatrix} 2 & -1 \\ -5 & 6 \end{bmatrix} = \begin{bmatrix} \frac{2}{7} & -\frac{1}{7} \\ -\frac{5}{7} & \frac{6}{7} \end{bmatrix}$$

We leave it for you to confirm that $AA^{-1} = A^{-1}A = I$.

(b) The matrix is not invertible since $\det(A) = (-1)(-6) - (2)(3) = 0$.

Example 2:

Consider the matrices

$$A = \begin{bmatrix} 1 & 2 \\ 1 & 3 \end{bmatrix}, \quad B = \begin{bmatrix} 3 & 2 \\ 2 & 2 \end{bmatrix}$$

We leave it for you to show that

$$AB = \begin{bmatrix} 7 & 6 \\ 9 & 8 \end{bmatrix}, \quad (AB)^{-1} = \begin{bmatrix} 4 & -3 \\ -\frac{9}{2} & \frac{7}{2} \end{bmatrix}$$

and also that

$$A^{-1} = \begin{bmatrix} 3 & -2 \\ -1 & 1 \end{bmatrix}, \quad B^{-1} = \begin{bmatrix} 1 & -1 \\ -1 & \frac{3}{2} \end{bmatrix}, \quad B^{-1}A^{-1} = \begin{bmatrix} 1 & -1 \\ -1 & \frac{3}{2} \end{bmatrix} \begin{bmatrix} 3 & -2 \\ -1 & 1 \end{bmatrix} = \begin{bmatrix} 4 & -3 \\ -\frac{9}{2} & \frac{7}{2} \end{bmatrix}$$

Remark:

If A is an invertible matrix, then A^T is also invertible and

$$(A^T)^{-1} = (A^{-1})^T$$

For a matrix of order higher than 2×2 we use the inversion algorithm to find its inverse (based on steps used for the reduced row echelon form).

1.5 Inversion Algorithm

To find the inverse of an invertible matrix A , find a sequence of elementary row operations that reduces A to the identity and then perform that same sequence of operations on I_n to obtain A^{-1} .

Example 4:

Find the inverse of

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 5 & 3 \\ 1 & 0 & 8 \end{bmatrix}$$

Solution We want to reduce A to the identity matrix by row operations and simultaneously apply these operations to I to produce A^{-1} . To accomplish this we will adjoin the identity matrix to the right side of A , thereby producing a partitioned matrix of the form

$$[A \mid I]$$

Then we will apply row operations to this matrix until the left side is reduced to I ; these operations will convert the right side to A^{-1} , so the final matrix will have the form

$$[I \mid A^{-1}]$$

The computations are as follows:

$$\begin{array}{l} \left[\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 2 & 5 & 3 & 0 & 1 & 0 \\ 1 & 0 & 8 & 0 & 0 & 1 \end{array} \right] \\ \left[\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 0 & 1 & -3 & -2 & 1 & 0 \\ 0 & -2 & 5 & -1 & 0 & 1 \end{array} \right] \leftarrow \begin{array}{l} \text{We added } -2 \text{ times the first} \\ \text{row to the second and } -1 \text{ times} \\ \text{the first row to the third.} \end{array} \\ \left[\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 0 & 1 & -3 & -2 & 1 & 0 \\ 0 & 0 & -1 & -5 & 2 & 1 \end{array} \right] \leftarrow \begin{array}{l} \text{We added 2 times the} \\ \text{second row to the third.} \end{array} \\ \left[\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 0 & 1 & -3 & -2 & 1 & 0 \\ 0 & 0 & 1 & 5 & -2 & -1 \end{array} \right] \leftarrow \begin{array}{l} \text{We multiplied the third} \\ \text{row by } -1. \end{array} \end{array}$$

$$\left[\begin{array}{ccc|ccc} 1 & 2 & 0 & -14 & 6 & 3 \\ 0 & 1 & 0 & 13 & -5 & -3 \\ 0 & 0 & 1 & 5 & -2 & -1 \end{array} \right] \leftarrow \begin{array}{l} \text{We added 3 times the third} \\ \text{row to the second and } -3 \text{ times} \\ \text{the third row to the first.} \end{array}$$

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 0 & -40 & 16 & 9 \\ 0 & 1 & 0 & 13 & -5 & -3 \\ 0 & 0 & 1 & 5 & -2 & -1 \end{array} \right] \leftarrow \begin{array}{l} \text{We added } -2 \text{ times the} \\ \text{second row to the first.} \end{array}$$

$$A^{-1} = \begin{bmatrix} -40 & 16 & 9 \\ 13 & -5 & -3 \\ 5 & -2 & -1 \end{bmatrix}$$

Example 5:

Consider the matrix

$$A = \begin{bmatrix} 1 & 6 & 4 \\ 2 & 4 & -1 \\ -1 & 2 & 5 \end{bmatrix}$$

$$\left[\begin{array}{ccc|ccc} 1 & 6 & 4 & 1 & 0 & 0 \\ 2 & 4 & -1 & 0 & 1 & 0 \\ -1 & 2 & 5 & 0 & 0 & 1 \end{array} \right]$$

$$\left[\begin{array}{ccc|ccc} 1 & 6 & 4 & 1 & 0 & 0 \\ 0 & -8 & -9 & -2 & 1 & 0 \\ 0 & 8 & 9 & 1 & 0 & 1 \end{array} \right] \leftarrow \begin{array}{l} \text{We added } -2 \text{ times the first} \\ \text{row to the second and added} \\ \text{the first row to the third.} \end{array}$$

$$\left[\begin{array}{ccc|ccc} 1 & 6 & 4 & 1 & 0 & 0 \\ 0 & -8 & -9 & -2 & 1 & 0 \\ 0 & 0 & 0 & -1 & 1 & 1 \end{array} \right] \leftarrow \begin{array}{l} \text{We added the} \\ \text{second row to} \\ \text{the third.} \end{array}$$

Since we have obtained a row of zeros on the left side, A is not invertible.

Work to do

Q1. Determine whether the statement is true or false, and justify your answer.

- (i) For all square matrices A and B of the same size
 $(A + B)^2 = A^2 + 2AB + B^2$.
- (ii) The product of two elementary matrices of the same size must be an elementary matrix.
- (iii) If A is an $n \times n$ matrix that is not invertible, then the linear system $A\mathbf{x} = \mathbf{0}$ has infinitely many solutions.

(iv) It is impossible for a linear system of linear equations to have exactly two solutions.

(v) If A and B are invertible matrices of the same size, then AB is invertible and $(AB)^{-1} = A^{-1}B^{-1}$.

(vi) Every elementary matrix is invertible.

(vii) If A is invertible and a multiple of the first row of A is added to the second row, then the resulting matrix is invertible.

(viii) If the linear system $A\mathbf{x} = \mathbf{b}$ has a unique solution, then the linear system $A\mathbf{x} = \mathbf{c}$ also must have a unique solution.

Q2. Use the inversion algorithm to find the inverses of the given matrices, if exist.

$$(i) \begin{bmatrix} 1 & 2 & -3 \\ 1 & -2 & 1 \\ 5 & -2 & -3 \end{bmatrix}$$

$$(ii) \begin{bmatrix} 0 & 2 & 3 \\ 1 & 1 & 1 \\ 5 & 5 & 1 \end{bmatrix}$$

$$(iii) \begin{bmatrix} 2 & 6 & 6 \\ 2 & 7 & 6 \\ 2 & 7 & 7 \end{bmatrix},$$

$$(iv) \begin{bmatrix} -1 & 3 & -4 \\ 2 & 4 & 1 \\ -4 & 2 & -9 \end{bmatrix},$$

$$(v) \begin{bmatrix} 2 & 5 & 5 \\ -1 & -1 & 0 \\ 2 & 4 & 3 \end{bmatrix}$$

Q3. Find all values of c , if any, for which the given matrix is invertible.

$$(i) \begin{bmatrix} c & c & 1 \\ 1 & 1 & c \\ 0 & 1 & c \end{bmatrix}$$

$$(ii) \begin{bmatrix} c & c & -1 \\ 1 & 1 & 2c \\ 0 & 1 & c \end{bmatrix}$$

$$(iii) \begin{bmatrix} c & c & c \\ 1 & c & c \\ 1 & 1 & c \end{bmatrix}$$

Q4. Find $\det A$ and $\det B$ using row or column of your choice and using reduced matrix.

$$A = \begin{bmatrix} -3 & 0 & 7 \\ 2 & 5 & 1 \\ -1 & 0 & 5 \end{bmatrix}, \quad B = \begin{bmatrix} 3 & 3 & 0 & 5 \\ 2 & 2 & 0 & -2 \\ 4 & 1 & -3 & 0 \\ 2 & 10 & 3 & 2 \end{bmatrix}$$

$$\text{Q5. If } \begin{vmatrix} a & b & c \\ d & e & f \\ g & h & i \end{vmatrix} = -6, \text{ then } \begin{vmatrix} 3a & 3b & 3c \\ -d & -e & -f \\ 4g & 4h & 4i \end{vmatrix} = ?$$